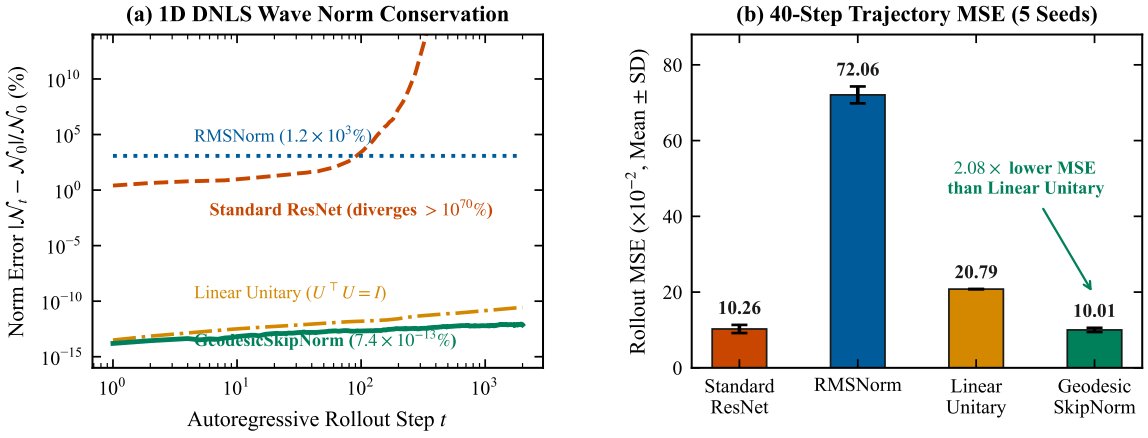

(a) 1D DNLS Wave Norm Conservation
Norm Error $|\mathcal{N}_t - \mathcal{N}_0|/\mathcal{N}_0$ (%)
10^{10}
10^{5}
10^{0}
10^{-5}
10^{-10}
10^{-15}
RMSNorm (1.2×10^{3}%)
Standard ResNet (diverges $> 10^{70}$%)
Linear Unitary ($U^\top U = I$)
GeodesicSkipNorm (7.4×10^{-13}%)
10^{0}
10^{1}
10^{2}
10^{3}
Autoregressive Rollout Step t
(b) 40-Step Trajectory MSE (5 Seeds)
Rollout MSE ($\times 10^{-2}$, Mean ± SD)
0
20
40
60
80
72.06
10.26
20.79
10.01
2.08× lower MSE
than Linear Unitary
Standard ResNet
RMSNorm
Linear Unitary
Geodesic SkipNorm